Mean Value Theorems: How Local Slopes Control an Interval

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A derivative is information at one point, whereas an estimate such as

$$|f(b) - f(a)| \le M|b - a|$$

compares two distinct points. The mean value theorem explains how to pass between these scales. We follow the classical route through extrema, Rolle's theorem, and its two-function extension, as developed in Rudin's *Principles of Mathematical Analysis*, Chapter 5, and the MIT 18.100B real-analysis lectures.[1]

§1.1 Why a compact interval enters the proof

Suppose $h : [a, b] \to \mathbb{R}$ is continuous and $h(a) = h(b)$. The extreme value theorem guarantees that h attains a maximum and a minimum. If both extrema occur only at endpoints, their common endpoint value makes h constant. Otherwise one extremum occurs at an interior point c. Provided h is differentiable there, the two one-sided difference quotients have opposite allowable signs, so $h'(c) = 0$.

Theorem 1.1.1 (Rolle)
If h is continuous on $[a, b]$, differentiable on (a, b), and $h(a) = h(b)$, then $h'(c) = 0$ for some $c \in (a, b)$.

The hypotheses cannot be omitted casually. For $h(x) = |x|$ on $[-1, 1]$, the endpoints agree but the sole interior minimum occurs at a point where the derivative does not exist. If continuity fails at an endpoint, the extreme value argument also breaks.

§1.2 Removing the secant line

Theorem 1.2.1 (Mean value theorem)
If f is continuous on $[a, b]$ and differentiable on (a, b), then there is $c \in (a, b)$ with

$$f(b) - f(a) = f'(c)(b - a).$$

Proof. The affine interpolant between the endpoint values is

$$\ell(x) = f(a) + \frac{f(b) - f(a)}{b - a}(x - a).$$

The difference $h = f - \ell$ satisfies $h(a) = h(b) = 0$. Rolle's theorem supplies c with

$$0 = h'(c) = f'(c) - \frac{f(b) - f(a)}{b - a}.$$

□

[1] https://ocw.mit.edu/courses/18-100b-real-analysis-spring-2025/pages/lecture-notes/

This proof suggests the correct geometric operation: subtract the simplest function matching the boundary data. For example, $f(x) = x^2$ on $[1, 3]$ has secant slope 4; the equality $2c = 4$ identifies $c = 2$.

§1.3 The two-function theorem and a nonzero denominator

Theorem 1.3.1 (Cauchy's mean value theorem)

Suppose f, g are continuous on $[a, b]$ and differentiable on (a, b). Then some $c \in (a, b)$ satisfies

$$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c).$$

Proof. Put $A = f(b) - f(a)$, $B = g(b) - g(a)$, and set

$$h(x) = Bf(x) - Ag(x).$$

Its endpoint difference is $BA - AB = 0$. Applying Rolle to h yields $Bf'(c) - Ag'(c) = 0$. □

The cross-multiplied identity is the theorem's basic form. To write a ratio one must justify the divisions. If $g' \neq 0$ throughout (a, b), Rolle applied to g forces $g(b) \neq g(a)$, and Cauchy's identity becomes

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}.$$

Without this extra condition, a quotient formulation may be meaningless although the cross-multiplied theorem remains true.

§1.4 Where l'Hopital's rule comes from

Consider the right-hand $0/0$ situation. Assume f, g are continuous on $[a, a + \delta)$, differentiable on $(a, a + \delta)$, with

$$f(a) = g(a) = 0, \qquad g'(x) \neq 0 \quad (a < x < a + \delta).$$

For every $x > a$, Cauchy's theorem on $[a, x]$ supplies $c_x \in (a, x)$ such that

$$\frac{f(x)}{g(x)} = \frac{f'(c_x)}{g'(c_x)}.$$

Consequently, if $f'(t)/g'(t) \to L$ as $t \downarrow a$, then $f(x)/g(x) \to L$, because $c_x \downarrow a$ in the sense that $a < c_x < x \to a$.

Example 1.4.1

Let $f(x) = \sin x - x$ and $g(x) = x^3$, with $a = 0$. Direct substitution gives $0/0$, but

$$\frac{f'(x)}{g'(x)} = \frac{\cos x - 1}{3x^2} \longrightarrow -\frac{1}{6}.$$

The derivative ratio here still uses the familiar limit $(1 - \cos x)/x^2 \to \frac{1}{2}$; it does not license a circular application of l'Hopital to itself.

The theorem is a statement about limits under explicit hypotheses. Taking derivatives of numerator and denominator is not a general algebraic operation on fractions.

§1.5 Derivative bounds give global estimates

Proposition 1.5.1 (Lipschitz estimate)

If f is continuous on an interval and differentiable in its interior, with $|f'(x)| \le M$, then

$$|f(x) - f(y)| \le M|x - y|$$

for all x, y in the interval.

Proof. Apply the mean value theorem to the subinterval with endpoints x, y, then take absolute values. □

If $f' = 0$ throughout an interval, this forces f to be constant. If $f' \ge 0$, the same reasoning shows f is nondecreasing. These are stronger conclusions than a picture of tangent lines: a pointwise differential inequality implies a global comparison.

The Lipschitz condition also enters uniqueness arguments for ordinary differential equations. When $|F(t, u) - F(t, v)| \le L|u - v|$, two solutions with the same initial value satisfy

$$e(t) := |u(t) - v(t)| \le L \int_{t_0}^{t} e(s)\, ds \qquad (t \ge t_0).$$

For $H(t) = \int_{t_0}^{t} e(s)\, ds$, one has $H(t_0) = 0$ and $H'(t) \le LH(t)$. Therefore $(e^{-Lt}H(t))' \le 0$. Since $H \ge 0$, it follows that $H = 0$ and the solutions coincide. This is the elementary core of the Gronwall argument.

§1.6 Continuity versus uniform continuity

Continuity at a point x allows the δ in the ε–δ definition to depend on x. Uniform continuity asks for the same $\delta(\varepsilon)$ across the domain.

Theorem 1.6.1 (Heine–Cantor)

A continuous function from a compact metric space to a metric space is uniformly continuous.

Proof. Suppose otherwise. There is $\varepsilon_0 > 0$ and sequences x_n, y_n with

$$d(x_n, y_n) < 1/n, \qquad d(f(x_n), f(y_n)) \ge \varepsilon_0.$$

By compactness, a subsequence x_{n_j} converges to some x. The triangle inequality gives $y_{n_j} \to x$ as well. Continuity at x forces both image subsequences to converge to $f(x)$, a contradiction. □

Compactness matters. The function $x \mapsto x^2$ on $\mathbb{R}$ is continuous but not uniformly continuous: take $x_n = n$, $y_n = n + 1/n$. The inputs approach one another, while the output difference tends to 2.

§1.7 Taylor's formula as an error statement

For $f \in C^2$ between a and x, Taylor's theorem gives some ξ between them such that

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(\xi)}{2}(x-a)^2.$$

In particular, if $|f''| \le K$, the error of the tangent-line approximation satisfies

$$|f(x) - f(a) - f'(a)(x-a)| \le \frac{K}{2}|x-a|^2.$$

The logical progression is now visible: extrema force a zero derivative; removing a secant produces a mean slope; bounding derivatives controls finite differences; controlling a higher derivative quantifies the error of a local polynomial approximation.